Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Jiajun He, Petros Sofronis	Email	jiajunhe@illinois.edu, sofronis@illinois.edu
Institution / Department	Department of Mechanical Science and Engineering, University of Illinois Urbana-Champaign			
Address	1206 W Green St, Urbana, Illinois 61801, USA			

2. Technology Overview

Title of Proposed Technology	Integrated Biogas Upgrading and Carbon Capture Platform for Clean Hydrogen Production from Agricultural Waste			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Mechanical・Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	*Hydrogen*	*Metal Materials*	*Materials Analysis/Evaluation Technologies*
Concept of Proposed Technology	The proposed technology is an integrated biogas upgrading platform that converts agricultural and waste-derived biogas into clean hydrogen while simultaneously capturing CO_2. The system combines high-temperature steam reforming and water–gas shift reactions with downstream CO_2 capture to enable low-carbon hydrogen production suitable for decentralized and industrial-scale applications. Beyond process integration, the technology incorporates rigorous techno-economic analysis (TEA) and life cycle assessment (LCA) to quantify cost, efficiency, and environmental performance, providing clear guidance for commercialization and deployment. A unique feature of the proposed approach is the explicit consideration of metal material integrity under high-temperature hydrogen environments. The technology evaluates hydrogen–materials interactions, including creep and degradation mechanisms, to ensure long-term reliability and safe operation of reformers, heat exchangers, and associated components. This coupling of system-level optimization with materials-level reliability analysis enables robust design and cost assessment of hydrogen production infrastructure. Potential uses include farm-scale and regional biogas upgrading, hydrogen supply for fuel cells and			

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	industrial processes, and deployment in distributed energy systems. The technology is particularly relevant to hydrogen infrastructure, carbon-neutral energy systems, and advanced manufacturing sectors seeking reliable, scalable, and sustainable hydrogen solutions.
Rationale for Joint R&D Project	The transition to a hydrogen-based energy system requires technologies that are not only low-carbon and cost-effective, but also reliable under high-temperature hydrogen operating conditions. While biogas-derived hydrogen offers a sustainable and promising pathway by utilizing agricultural and waste resources, its practical deployment is limited by system integration challenges, uncertain economics, and long-term material degradation in hydrogen-rich and high-temperature environments. Addressing these barriers requires coordinated advances across process design, sustainability assessment, and materials reliability. This joint R&D project is needed to bridge the gap between laboratory-scale hydrogen production concepts and industrially deployable systems. Collaboration between the UIUC research team and Korean small and medium-sized enterprises (SMEs) enables the integration of advanced modeling, techno-economic and life-cycle assessment capabilities with industrial design, manufacturing, and validation expertise. Korean SMEs bring critical knowledge of system fabrication, operational constraints, and market needs, while the UIUC team contributes fundamental understanding of high-temperature hydrogen–materials interactions and system-level optimization. Through this collaboration, the project will accelerate technology maturation, reduce technical and economic risks, and support the development of scalable, durable hydrogen and CO_2 capture systems aligned with Korea's carbon neutrality and hydrogen economy goals.
Keywords	1. Biogas-to-Hydrogen Conversion 2. Integrated CO_2 Capture 3. High-Temperature Hydrogen Materials Integrity 4. Techno-Economic and Life Cycle Assessment (TEA-LCA) 5. Distributed Clean Energy Systems

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

Existing Technologies and Capabilities

The research team currently possesses strong, complementary capabilities spanning energy systems modeling, sustainability assessment, and materials reliability in hydrogen environments. On the system level, the team has established modeling frameworks for thermochemical conversion processes, including reforming and gas-phase reaction networks, coupled with rigorous techno-economic analysis (TEA) and life cycle assessment (LCA). These tools enable quantitative evaluation of cost, efficiency, and greenhouse gas emissions for integrated energy and carbon capture systems. Prior work has demonstrated the ability to link experimentally validated material performance data with process-level thermodynamic and economic models.[1-4]

On the materials side, the team has extensive expertise in high-temperature hydrogen–materials interactions, including creep, hydrogen attack, and degradation mechanisms in structural metals.[5-7] The team's existing modeling and experimental methodologies allow prediction of material performance, damage evolution, and lifetime under hydrogen exposure, providing design-relevant criteria for safe operation of hydrogen-facing components. High-temperature and high-pressure hydrogen experiments, when required, will be conducted at the International Institute for Carbon-Neutral Energy Research (I^2CNER) in collaboration with Kyushu University, which offers specialized facilities and expertise for such testing.

Planned Technology Development through this R&D Project

Through this joint R&D project, the team will integrate these existing capabilities into a unified biogas-to-hydrogen platform with embedded CO_2 capture and materials reliability considerations. New developments will include system-level integration of reforming, water–gas shift, and CO_2 capture processes; coupling of TEA/LCA with hydrogen infrastructure design constraints; and translation of hydrogen–materials interaction models into engineering guidelines for reformers, heat exchangers, and related components. This integration will advance the technology toward a deployable, reliability-informed hydrogen production system suitable for SME-led commercialization.

R&D Target Specifications

The joint R&D project aims to establish quantitative performance benchmarks for an integrated biogas-to-hydrogen system with embedded CO_2 capture and materials reliability considerations. Target specifications are defined at the system, sustainability, and materials integrity levels to ensure technical feasibility and industrial relevance.

System-Level Performance Targets:

- Hydrogen purity: ≥ 99.9 vol% H_2 (after reforming, water gas shift, and purification)
- Overall hydrogen yield: ≥ 65–75% (based on biogas energy content, LHV basis)
- CO_2 capture efficiency: ≥ 90% of CO_2 in the reformer/water gas shift outlet stream
- System thermal efficiency: ≥ 60% (biogas-to-hydrogen, LHV basis)

Techno-Economic and Environmental Targets:

- Levelized cost of hydrogen (LCOH): ≤ 4–6 USD/kg-H_2, depending on feedstock and scale
- Net life-cycle CO_2 emissions: ≤ 1.0 kg CO_2-eq/kg-H_2, with pathways toward net-negative emissions when agricultural waste credits are included
- CO_2 payback time: ≤ 1–2 years when integrated with CO_2 capture systems

Materials Integrity and Reliability Targets:

- Operating temperature range: 700–900 °C for hydrogen-facing components
- Predicted component lifetime: ≥ 20,000–40,000 hours under high-temperature, high-pressure hydrogen exposure

	• Creep and hydrogen damage safety margin: ≥ 2× against predicted failure thresholds • Validated hydrogen–materials degradation models: Error ≤ 20% compared to experimental data. Benchmarks will also be developed using mechanism-based Nelson curves[6] against high-temperature hydrogen attack. These target specifications will guide system design, materials selection, and scale-up decisions, providing Korean SMEs with actionable performance benchmarks for pilot and commercial deployment.
R&D Plan	**Overall Research Approach** The project adopts a system-integrated, reliability-informed approach that combines process modeling, sustainability assessment, and materials integrity analysis. Biogas reforming, water–gas shift (WGS), and CO_2 capture are developed as an integrated system, while high-temperature hydrogen-materials interaction models are embedded to guide safe and durable component design. Continuous feedback between system performance, economics, environmental impact, and materials reliability ensures realistic scale-up pathways for industrial deployment. The project is structured in two stages aligned with the TIPA program: Phase 1 (6 months) focuses on joint planning, feasibility assessment, and system definition, while Phase 2 (2.5 years) executes integrated technology development, validation, and scale-up preparation. **Phase 1 (Seed Funding): Joint Planning and Feasibility Assessment (Months 0–6)** **Task P1-1: System Definition and Use-Case Alignment** • Define target applications (farm-scale, regional, industrial). • Identify SME-relevant operating constraints, manufacturing considerations, and performance targets. • Select representative biogas compositions and operating envelopes. Deliverable: Agreed system scope and performance targets. **Task P1-2: Preliminary Process and Integration Assessment** • Develop preliminary models for biogas reforming, WGS, and CO_2 capture integration. • Screen alternative configurations and operating conditions. • Identify key technical risks and integration challenges. Deliverable: Feasible system configurations and risk assessment. **Task P1-3: Initial TEA/LCA Scoping** • Conduct high-level TEA and LCA to identify cost and emissions drivers. • Define data needs and benchmarks for Phase 2 development. • Deliverable: Go/no-go metrics and Phase 2 R&D roadmap. **Phase 1 Milestone:** Jointly defined, technically feasible biogas-to-hydrogen system with CO_2 capture and a detailed Phase 2 Execution plan (TRL 3). **Phase 2 (2.5-Year Full Award Phase) : Full R&D Execution (Months 7–36)** **Task 2-1: Detailed Process Modeling and Optimization (Months 7–14)** • Develop detailed thermodynamic and kinetic models for reforming and WGS. • Optimize hydrogen yield, efficiency, and operating conditions. • Refine system integration strategies. Milestone: Optimized the hydrogen production process model. **Task 2-2: CO_2 Capture Integration and System Optimization (Months 12–20)** • Integrate and optimize CO_2 capture units.

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	• Minimize energy penalty while maintaining ≥90% capture efficiency. • Quantify effects on hydrogen purity and system efficiency. Milestone: Validated integrated hydrogen + CO_2 capture system. **Task 2-3: Techno-Economic and Life Cycle Assessment (Months 14–26)** • Perform detailed TEA to estimate hydrogen cost and scalability. • Conduct full LCA to quantify net CO_2 emissions and mitigation potential. • Perform sensitivity and scenario analyses relevant to SME deployment. Milestone: Cost and environmental performance benchmarks. **Task 2-4: High-Temperature Hydrogen–Materials Integrity Analysis (Months 18–30)** • Evaluate creep, hydrogen attack, and degradation mechanisms. • Translate mechanistic models into lifetime and safety criteria. • Link materials limits to system operating conditions. Milestone: Reliability-informed design guidelines for hydrogen-facing components. **Task 2-5: System Integration, Validation, and Scale-Up Guidance (Months 28–36)** • Integrate process, sustainability, and materials results. • Validate system performance against target specifications. • Develop scale-up and pilot-design guidance for Korean SMEs. Milestone: TRL 5 system-level validation and commercialization roadmap. **Expected Outcome** By the end of Phase 2, the project will deliver an integrated, reliability-informed biogas-to-hydrogen technology with quantified economic, environmental, and durability performance, enabling Korean SMEs to proceed confidently toward pilot-scale demonstration and commercialization.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The UIUC research team will lead the fundamental and system-level research required to develop an integrated, reliability-informed biogas-to-hydrogen technology with CO_2 capture. The team will provide core technical expertise in process modeling, sustainability assessment, and materials integrity under high-temperature and high-pressure hydrogen environments. Jiajun He will be primarily responsible for system-level analysis and sustainability-driven design. This includes development of thermodynamic and kinetic models for biogas steam reforming, water–gas shift reactions, and CO_2 capture integration, as well as comprehensive techno-economic analysis (TEA) and life cycle assessment (LCA). His role also includes defining performance benchmarks, identifying cost and emissions drivers, and translating system-level results into scalable design guidelines relevant to industrial deployment. Petros Sofronis will lead the materials integrity and reliability component of the project. His responsibilities include modeling and analysis of hydrogen–materials interactions at high temperature, including creep, hydrogen attack, and degradation mechanisms in structural materials used in hydrogen-facing components. He will develop lifetime prediction methodologies and safety margins that link operating conditions to material performance. Together, the UIUC team will integrate system performance, sustainability metrics, and materials reliability into unified design frameworks, provide technical reports and design guidelines, and support

	knowledge transfer to Korean SME partners throughout Phase 1 planning and Phase 2 execution.
Expected R&R of Korean SME Partner	The Korean SME partner is expected to play a central role in translating the proposed technology from a system-level design into an industrially relevant and commercially viable solution. The SME will contribute practical engineering expertise, manufacturing insight, and application-specific requirements that are essential for technology maturation. Specifically, the SME partner is expected to provide input on system design constraints, component specifications, and operational considerations for biogas reforming, CO_2 capture, and hydrogen-handling equipment. This includes contributions to hardware design concepts, materials selection, fabrication feasibility, and integration with existing industrial processes or energy infrastructure. The SME will support prototype development and pilot-scale validation, including testing under representative operating conditions and evaluation of system performance, durability, and safety. The SME partner is also expected to share relevant operational data, cost information, and market perspectives to inform techno-economic and life cycle assessments. Resource commitments may include access to testing facilities, engineering personnel, and in-kind support for system integration and validation activities. Through close collaboration with the UIUC research team, the SME partner will participate in joint planning, iterative design refinement, data exchange, and regular technical reviews, enabling efficient technology development and accelerating pathways toward commercialization and deployment in the Korean market.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to deliver a fully integrated and reliability-informed biogas-to-hydrogen technology that advances low-carbon hydrogen production while addressing key technical and economic barriers to deployment. Major technological outcomes include validated system designs that combine biogas steam reforming, water–gas shift reactions, and high-efficiency CO_2 capture, supported by comprehensive techno-economic analysis (TEA) and life cycle assessment (LCA). In parallel, the project will generate quantitative design criteria for hydrogen-facing components by incorporating high-temperature and high-pressure hydrogen–materials interaction and degradation mechanisms, enabling improved lifetime prediction and safer operation of critical equipment. From a commercialization standpoint, the project will provide Korean SMEs with actionable design frameworks, performance benchmarks, and cost targets that reduce uncertainty in scaling and deployment. The integration of system-level optimization with materials integrity analysis will lower technical risk, shorten development cycles, and facilitate transition from prototype validation to pilot-scale and early commercial systems. These outcomes are particularly relevant for SMEs engaged in hydrogen production, carbon capture technologies, pressure vessels, heat exchangers, and energy system integration. At a broader level, the project will support Korea's hydrogen economy and carbon neutrality objectives by enabling sustainable hydrogen production from agricultural and waste-derived resources. The technology promotes waste valorization, distributed energy systems, and durable hydrogen infrastructure, contributing to reduced greenhouse gas emissions, enhanced energy security, and new

	industrial opportunities within the Korean energy and manufacturing sectors.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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